

The Multifaceted Nature of Robotic Companionship when Presented as a Secondary Function

Hadas Erel

hadas.ere1@milab.idc.ac.il

Media Innovation Lab (miLAB), School of
Communications, IDC Herzliya
Herzliya, Israel

Adi Manor

adi.manor@milab.idc.ac.il

Media Innovation Lab (miLAB), School of
Communications, IDC Herzliya
Herzliya, Israel

Omer Dado

omer.dado@milab.idc.ac.il

Media Innovation Lab (miLAB), School of
Communications, IDC Herzliya
Herzliya, Israel

Oren Zuckerman

orenz@idc.ac.il

Media Innovation Lab (miLAB), School of
Communications, IDC Herzliya
Herzliya, Israel

ABSTRACT

Companion robots have been suggested as a promising technology for older adults who experience loneliness. However, healthy older adults commonly reject robots designed to be an "artificial friend". We follow the approach of "companionship as a secondary function", in which a non-humanoid robot is designed with a primary function that older adults perceive as appropriate, and a secondary function of companionship. In a Zoom-based exploratory need-study we unfold how older adults perceive the various aspects of a robot's "companionship" as a secondary function. Our qualitative analysis reveals several use cases that older adults find to be appropriate for their daily routine, and classify them into three high-level categories: companionship as "attentive to me", companionship as "looking after me", and companionship as "experiencing together with me". Our findings indicate that robot companionship, when designed as a secondary function, is perceived by older adults as a multifaceted social experience.

CCS CONCEPTS

• Human-centered computing → Empirical studies in HCI.

KEYWORDS

Non-humanoid robots, older adults, companionship robot, successful aging

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1 INTRODUCTION

In recent years, there is an increasing need for technologies designed specifically to support successful aging [7, 43, 46]. A growing population of older adults maintain relatively high-levels of physical and cognitive functioning [7, 33, 43] and have the potential to experience successful aging [7, 12]. Companion robots have been suggested as a potential technology for promoting successful aging, as these robots are designed to provide social experiences for older adults, with the goal of reducing challenges associated with loneliness [21, 25, 26, 35]. Most companion robots take on a humanoid or zoomorphic appearance (e.g. seal, cat, dog) and are typically presented as a "friend" with the ability to provide companionship [16]. Indeed, such companion robots were shown to be successful in reducing feelings of loneliness with clinical populations, including older adults who suffer from mild cognitive impairment or dementia [18, 31, 40]. However, healthy older adults commonly object to their "artificial companionship" and reject them [7, 38, 46]. This challenge should be addressed by designing companion robots that will be accepted by healthy older adults, increasing their potential to experience successful aging.

Prior studies have already mapped specific design aspects and preferences that are important when designing for healthy older adults [7, 13, 41]. These studies indicate that the robots' function and appearance are key factors contributing to the robot's acceptance [7, 13, 41, 46]. In terms of function, it was indicated that companionship is not considered as a valid primary function by healthy older adults [46] and that acceptance is associated with functions that older adults themselves perceive as meaningful [6, 46]. However, companionship and specifically social interaction is accepted by healthy older adults when it is presented as a secondary function [7, 46]. A recent lab study, introduced a robotic object with cognitive preservation as a primary function and companionship as a secondary function. Older adults accepted the robot's social secondary function and associated the robot's minimal gestures with feelings of being seen [46]. The specific use cases that older



Figure 1: A non-humanoid robot designed as an anagram word game with a primary function of cognitive preservation. The robot's social aspects are designed as a secondary function, using non-verbal gestures [46]

adults accept as appropriate for robotic companionship when implemented as a secondary function have not been mapped yet, and this is the focus on this exploratory study.

The robot's appearance is another important aspect associated with the robot's acceptance. Healthy older adults reject robots "pretending to be a friend" [7, 10, 16]. In some cases, older adults strongly object to humanoid or zoomorphic designs (i.e. human-like or animal-like), describing the design as not authentic and artificial [7]. Older adults commonly prefer robots with appearance that reflects their function (which cannot be companionship [7, 45, 46]). One option for addressing this preference is to design non-humanoid robotic objects with an appearance that serves their function [46]. Robotics objects are especially relevant, as they can perform various non-verbal gestures that humans perceive as social cues. Previous studies with such robots have indicated that it is possible to create a social experience despite the non-humanoid appearance and the inability to mimic human behavior [2, 46]. Participants attributed social intent even to minimal gestures, performed either by abstract robotic objects or by robots designed as everyday objects [2, 19, 37, 46]. Moreover, participants associate social interpretation with non-humanoid robots' gestures even in a non social context, for example when they are asked to report the physical direction of the robot's movement [8]. These studies suggest that with appropriate design consideration, it is possible to create a non-humanoid robot with an appearance that reflects its primary function, and with social gestures as a secondary function.

Companionship that is applied as a secondary function should be further explored as the robot is not directly designed to "be a friend". The challenge is to identify social functions older adults perceive as acceptable [23]. To overcome this challenge, we follow the approach of involving older adults in the design process [34]. It is suggested that consulting with older adults during the design process can highly benefit the fit and meaningfulness of the technology [24, 34]. Specifically regarding social functions, involving older adults in the decisions concerning a robot's social interactions was shown to lead to developing robots that are perceived by older adults as appropriate for their history and for their everyday experiences [42]. Moreover, consulting with older adults in the design

process was shown to assist in overcoming acceptance barriers, (e.g uneasiness with technology and feeling of stigmatization), and mitigating concerns relevant for this population [24].

In this work, we conducted a Zoom-based exploratory need-study with older adults to better understand which social use cases they find to be appropriate in the context of their daily routines. We used videos of a non-humanoid robot (used with permission), which has a primary function of a cognitive game and a secondary function of providing social experiences via non-verbal gestures. The exact same robot and gestures were validated in a prior lab study, in which the social gestures evoked a feeling of "being seen" by healthy older adults [46]. In this Zoom-based video study (conducted remotely due to the COVID-19 pandemic), we showed older adult participants three different videos of the robotic object in a home context, and asked them to suggest potentially appropriate and relevant social use cases for such a robot in their daily routines.

2 RELATED WORK

Most existing companion robots for older adults are specifically designed to provide companionship in an effort to reduce the negative influences of loneliness [32]. Companion robots typically take zoomorphic form and are designed to resemble pets. Leading examples for companion robots include Paro, a robotic seal; Huggable, a robotic teddy bear; and Aibo, a robotic dog [5, 20, 39]. Studies conducted with older adults suffering from dementia, mild cognitive impairment, and depression have shown positive influences when interaction with animal-like companion robots. The interaction with such companion robots led to consistent success in reducing feelings of loneliness, improving mood, reducing stress, increasing well-being, and reducing depression symptoms [1, 18, 44]. When tested with healthy older adults, companion robots are commonly rejected. Participants perceive these robots as unacceptable for interaction and define them as "unproductive entities" [7, 9].

For example, Lazar et al. (2016) [22] examined perceptions by older adults with regards to robotic pets and the ways in which they expressed interest or disinterest in interacting with them. The older adults did not perceive the robotic pets as an appropriate solution for alleviating social isolation. They stated that the robots may be useful for others, who are older and physically frail. The participants strongly rejected the idea of a robotic pet as a solution for loneliness. They did welcome the idea of having a robot to fill certain social needs that were not currently satisfied or to enhance social interactions with others. However, participants did not want a robot that solely contributed functional assistance. Instead, they wanted to care for assistive robots and to build a relationship with them.

An example for companionship by an abstract non-humanoid robot was introduced by Zuckerman et al. (2020) [46]. The non-humanoid robot was designed as an anagram word game with a primary function of cognitive preservation. The robot's social aspects were designed as a secondary function, using non-verbal gestures. Healthy older adults who interacted with the robot associated the robot's gestures with a social interaction related to the feelings of "being seen", known to reduce loneliness. The older adults perceived the primary function of cognitive preservation as a valid function for a robot and this increased their willingness

to accept the robot. In this study, we use the same robotic object [46] to explore healthy older adults' ideas for use cases they find as relevant for social interactions in their home context.

3 METHOD

3.1 Participants

The study involved 12 participants (7 males and 5 females; average age = 70.833, SD = 4.428). All participants were healthy older adults who live independently at home, reported active lifestyles, and were technically capable of using Zoom. Older adults were recruited from a database of older adults, who participated in a series of on-campus lectures and volunteered to participate in the lab's research projects. None of the older adults participated in academic studies prior to this research. Participants signed a consent form which was sent to them by email prior to the experiment. As appreciation for their participation, the participants were invited to an online lecture by the lab staff.

3.2 Experimental settings

Interviews were scheduled over the phone and participants were informed that the interview would be conducted via Zoom. The researcher verified that they had used Zoom in the past, and that they know how to operate it. At the end of the conversation, the researcher sent a Zoom link and the consent form to the participant's email. On the day of the experiment, the researcher and the participant opened the Zoom conversation at the designated time and the researcher made sure that the camera and sound were working properly, that the internet connection was stable and that the session was being recorded. During the interview, the researcher used the "Share Screen" function when presenting videos that were an integral part of the interview.

3.3 The robotic object

The robotic object is a non-humanoid robot, designed as an anagram word game (see Figure 1), capable of performing non-verbal gestures. The robot is designed with four rotating pillars, each representing a set of letters (or a blank space), and the distance between the pillars allows for the letters to be perceived as a whole word. In order to play the anagram game, users manipulate (rotate) the pillars until a word is created. The non-verbal gestures are designed by an animator using a 3D model of the robotic object, which includes the exact morphology and mechanism limitations. Previous work with the robotic object indicated that despite its abstract non-humanoid appearance, its gestures are interpreted by older adults as indicating social cues associated with a feeling of "being seen" [46].

3.4 Interview

Participants' ideas and thoughts about situations relevant for the non-humanoid robot were assessed using a semi-structured interview that lasted approximately 45 minutes. The interview's flexibility provides the opportunity to reach relevant information while remaining responsive to the participant [27]. In this study, the researcher constructed 10 questions in advance, including: "If you had such a robotic object at your home, how would you use it?";



Figure 2: A Zoom based exploratory need-study with older adults, conducted via Zoom

"What are the movements that the robotic object should perform?"; "Which situations are relevant for using the robot?"; "Why do you think this situation is relevant?"; "where would you place the robot in your home? why?". The questions were presented to the participant and additional questions were added based on the participants' responses (see Figure 2).

3.5 Procedure

The Zoom interviews were scheduled with the participants over the phone at a date and time convenient for them. Informed consent forms were sent by mail and participants signed the consent forms and sent them back to the researcher prior to the interview. On the day of the study the researcher contacted the participant and initiated the Zoom conversation.

The study began with a short conversation regarding the participant's daily routine. The purpose of the conversation was to create a relaxed atmosphere and to confirm orientation to time and space. All participants demonstrated clear orientation to time and space. The conversation was followed by the presentation of the robotic object using a photo and an explanation of the robotic object's appearance and mechanical components. The participants then saw a pre-recorded introduction video (see Figure 3) that was played using the "Share Screen" Zoom option. In the video, the robotic object was presented as an anagram word game. A person is constructing words using the robotic object's pillars and once a word is constructed the robot performs a "bow" gesture as a feedback (pillars moving backward and forward). After verifying that the participant understood the robot's primary function (i.e. an anagram word game), participants were told that we are developing the robotic object and that we are interested in their personal impressions, general attitudes, ideas and preferences. The researcher explained that there is no correct or incorrect answer and that their insights will help us improve the design of the robot's adaptation to the context of a home. The researcher further emphasized that we are

interested in their genuine opinion and reflection. This explanation was followed by the presentation of two additional videos showing the robot performing gestures that were perceived as social cues in a previous study [46].

In video 1, the robot performed a welcoming gesture, whenever a person walked into the room. The person enters facing the robot that is placed on a table. As the person enters the room, the robot performs a gesture designed as 4 pillars moving forward gradually. In video 2, a person is sitting next to a table where the robot is placed, the person takes the newspaper and the robot performs a follow gesture, all pillars move gradually at the same time towards the direction of the person.



Figure 3: The older adult is presented with videos showing an example of the robotic object's social gestures, and suggests use cases relevant for a home context

The researcher verified that the participant understood the scene presented in the video and started the semi-structured interview. Participants were asked to suggest scenarios for using the robotic object in their home. They were encouraged to share where they would place the robot in their home and how they would use it. Each scenario was discussed separately and participants elaborated on situations that they find relevant for the robot. The interview ended by thanking the participants for their help and asking for their permission to contact them again in the future.

4 ANALYSIS

The semi-structured interviews were transcribed and analyzed using the thematic coding method [3, 11]. The analysis included five stages: (1) Interviews were transcribed and half of the interviews were read several times by two coders to develop a general understanding of the data before the coding process began; (2) Initial themes were identified, presented to a third researcher, and discussed in-depth until inconsistencies were resolved; (3) A list of mutually-agreed themes was defined; (4) The raters used these themes to analyze half of the data independently, verifying inter-rater reliability ($\kappa=89\%$); (5) The two coders analyzed the rest of the data.

5 FINDINGS

A total of 56 quotes were associated with use cases and situations that the older adults found as appropriate for the robot in their daily routine at home (a quote was defined as a sentence including its continuation if participants further explained the exact same topic). The analysis revealed six themes grouped into three high-level categories: companionship as "attentive to me" ("Responsive motivator" and "Being seen" themes), companionship as "looking after me" ("Looking out", "Indicating that something is happening", and

"Reminder" themes), and companionship as "experiencing together with me" ("Co-experience" theme).

5.1 Attentive to me

Responsive motivator: 15/56 of the quotes were associated with the "Responsive motivator" theme. Participants thought of the robotic object as a companion that will encourage and motivate them to do things they do not like to do but find important. They suggested that the robot will provide both positive and negative feedback, which will encourage them to perform wanted behaviors and will reduce their tendency to perform unwanted behaviors. Wanted behaviors included cooking, exercising, cleaning, and cognitive activities such as Sudoku: "When I finish exercising, I would like him to cheer for me" (p.4, F, 78); "When I finish solving a Sudoku, the robot will show me that he is happy for me" (p.2, F, 64). The negative behaviors included watching television for long periods of time, using the phone for long periods of time, and wasting time performing actions that are not meaningful: "The robot will signal that he wants me to stop using my phone, because I used it for a long time". (p.3, F, 64).

Being seen: 8/56 of the quotes were associated with the "Being seen" theme. Participants suggested that the robot should make them feel as if someone is paying attention to them, that they are being noticed. They stated that the robot should greet them and acknowledge them, for example, "When I wake up in the morning, I would like him to notice and greet me good morning" (p.9, M, 69). Others wanted the robot to notice their emotional state: "When I feel down and moody, I would like him to respond by showing me that he saw that I am in a bad mood" (p.1, M, 72). In some cases, participants just wanted the robot to notice them and respond to their presence when they are next to it: "I want him to notice me while I'm performing my daily activities" (p.10, M, 74).

5.2 Looking after me

Looking out: 7/56 of the quotes were associated with the "Looking out" theme. Participants suggested that the robotic object will serve as an alarm in problematic situations: "The robotic object will signal that there is a danger, for example a thief or a fire". In other cases they suggested the robotic object should verify that they are fine: "Before I leave my house, I want him to signal if I am dressed inappropriately". (p.11, M, 71).

Indicating that something is happening: 10/56 of the quotes were associated with the "Indicating that something is happening" theme. Participants suggested that the robot will signal that their attention is required in the house. In most cases, they thought of the gestures as indicating different events, for example, a knock on the door, a phone ringing, or even more intimate events such as their grandchildren waking up and crying: "When my grandchildren stay with me I want it to signal if they wake up and cry". (p.6, F, 70).

Reminder: 6/56 of the quotes were associated with the "Reminder" theme. In some cases participants wanted the robotic object to be a reminder for actions and events: "When i'm busy and I forget to do things that I needed to do, I would like him to remind me to check what I missed" (p.6, F, 70). In other cases they proposed that the robot could signal that a specific amount of time has elapsed "When

my grandchildren watch television, I would like him to signal to me that their T.V. watching time is over" (p.3, M, 72).

5.3 Experiencing together with me

Co-experience: 10/56 of the quotes were associated with the "Co-experience" theme. Some participants suggested that the robot will share experiences with them: "When music is played in the room, I would want it to move with the music's beat" (p.10, M, 74); "When I'm watching a soccer game, I would like him to join my responses, to be happy when there is a goal and sad when a goal is missed" (p.1, M, 72).

6 DISCUSSION

In an exploratory need-study with older adults we identified various contexts of use that older adults find appropriate for a robotic object that performs non-verbal social gestures. The robot's social features were implemented as its secondary function. Previous studies indicated that robotic objects can lead to social experiences, however, they did not explore older adults' perception of relevant use-cases for these social experiences. The study was conducted remotely via Zoom, using videos of the robotic object performing different gestures. Based on prior studies, social companionship was implemented as a secondary function of a non-humanoid robot. The robot was not designed to "be a friend" [17] as artificial friendship is commonly rejected by healthy older adults [7]. Our findings indicate that companionship as a secondary function is perceived by older adults as a multifaceted experience that expands the notion of robotic social companionship, and may include social aspects relevant for the daily routines of older adults at home.

The qualitative analysis revealed three high-level categories: companionship as "attentive to me", companionship as "looking after me", and companionship as "experiencing together with me". The first category indicated that older adults perceived the robotic object as an appropriate companion for helping them achieve their goals by encouraging them when performing behaviors that they perceive as positive and signaling that they are performing behaviors that they perceive as negative. They also perceived the robot as having the potential to enhance the feeling of being noticed and seen, despite the robot's non-humanoid appearance (supporting findings by Zuckerman et al., 2020). The companionship as "looking after me" category indicated that older adults thought of the robotic object as a way to relieve their worries by making sure they do not miss significant events that may be important for their daily routine. The types of events participants mentioned ranged from alarming incidents (someone entering their home without permission), to events that they should attend (someone calling them), and prospective happenings that they should not forget (a meeting listed in their calendar). Companionship as "experiencing together with me" category indicated that older adults believed the robotic object could be a responsive partner for joining their leisure activities. This idea is supported by previous studies with a non-humanoid robot that indicated how having a robotic object join a music listening experience could enhance the experience [15].

In sum, older adults suggested various social use cases as valid and appropriate secondary functions for a non-humanoid robotic

object. The use cases that participants suggested were all in-line with factors mentioned in previous studies, which evaluated older adults' perception of companion robots [7, 23, 28]. The use cases did not involve a robot replacing interactions with other humans or a robot that performs tasks instead of older adults. In addition the cases did not compromise older adults' sense of control and independence. Instead, the use cases involved interactions where the robot is attentive to them, looks after them, and experiences together with them. It is possible that the use-cases suggested by the older adults represent a broad perception of social use-cases that they find appropriate and relevant for the interaction with technology in general, implying that different types of technologies will be perceived as acceptable for such interactions (e.g. voice interfaces). This possibility should be tested in future studies.

7 CONCLUSION

In this exploratory need-study we showed that healthy older adults are able to associate a robotic object's social gestures with various use cases that are appropriate for their daily routine. The social gestures were presented as a secondary function of a non-humanoid robot. The older adults in our study suggested a variety of use cases, unfolding the notion of companionship into a multifaceted experience. The use cases represented social functions relevant for the older adults, including being attentive to them, looking after them, and joining their leisure activities. Future studies should explore these initial findings, analyzing the effect such social functions have on older adults when implemented as a secondary function, and the potential to enhance older adults' well-being and successful aging.

8 LIMITATIONS

This study has several limitations. As a qualitative study based on interviews, our interviewers could unknowingly influence an interviewee's responses [30]. To limit this effect, the interviewers were trained according to a detailed interview protocol and we increasing their awareness to this effect [30]. An additional known limitation concerns the "good subject effect" [29] and participants wish to provide pleasing answers. To limit this effect, participants were reassured that anything said is helpful and valuable. In addition, our choice of Zoom based interviews and video study rather than face-to-face interviews and live interaction with the robotic object may have influenced the findings. This choice was due to the limitations of running a study in times of COVID-19 pandemic, which was extremely challenging specifically with older adults. We note that video studies are a known method in human-robot interaction research. Disadvantages of this method include a less realistic experience [36], and the lack of direct, live interaction between the older adult and the robotic devices [4]. However it also allows for increased consistency [14, 45], and specifically in this case, to conduct an initial assessment with older adults under COVID-19 challenges.

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